

Complete Timeline

Additional Resources- <https://platform.claude.com/docs/en/build-with-claude/prompt-engineering/overview>

1. Week 1- Foundations of LLMs and RAG-

Rag explanation-

<https://colab.research.google.com/github/mrdbourke/simple-local-rag/blob/main/00-simple-local-rag.ipynb>

RAG paper-

<https://arxiv.org/pdf/2005.11401>

Practical Implementation task- Learn How to retrieve and Store research Paper(PDF) and make a Pipeline using RAG which basically use case of RAG

2. Week 2- Learn Lang- Graph and Need for Agentic Systems-

<https://huggingface.co/learn/agents-course/unit2/langgraph/introduction>

<https://docs.langchain.com/oss/python/langgraph/overview>

go through the paper- <https://arxiv.org/pdf/2210.03629>

Practical implementation task- Make a sub part of the problem ,basically make a agent that decomposes the problem statement into small parts and thinks what the INTER IIT PS is actually demanding

reranking basics- BGE reranker, sentence-transformers reranking

3. Week 3- Make a proper Agent Architecture-

Think of all the agents needed and tools needed and work on each of the tool for the whole week.

Think about citation traversal, How we will properly explore all the citations given in the paper

4. Week 4-Learn about the Critic Agent-

<https://arxiv.org/abs/2310.11511> go through this research paper and learn about the need of Critic and incorporate it in the pipeline

5. Week 5- learn About MCP

<https://modelcontextprotocol.io/docs/getting-started/intro> ,incorporate it into your project

6. Week 6-Learn what all improvements can be done and think what all can be included

evaluation metrics

also compare-

- naive RAG
- agentic RAG
- agentic RAG + reranking

7. Week 7- Do proper testing and think improvements, Learn about lang-Smith,Think about the costs and optimisation

- explainability/logging
- benchmark comparisons

8. Week 8- Try to deploy or give it a proper structure

